Nonfinite constructions in Germanic

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Abstract

This article provides a comprehensive descriptive overview of nonfinite constructions (NFCs) across the Germanic language family, highlighting both their structural unity and empirical diversity. The analysis focuses on empirical generalizations that correlate with typological features, particularly the basic word order within the verbal phrase. Following a definition of nonfinite forms as verbal predicates lacking person and number marking, the article surveys the morphological spectrum of NFCs. It then examines their relational properties by distinguishing between raising and control, as well as between non-sentential (coherent) and sentential (incoherent) configurations. Both construction types are discussed in greater detail. For non-sentential NFCs, the discussion focuses on the verb cluster property characteristic of Continental West Germanic languages, including variation in serialization patterns and morphosyntactic alternations such as the 'substitute infinitive' (IPP). For sentential NFCs, it reviews their syntactic functions and clause types, including specialized constructions such as root infinitivals.

Contents

1 Introduction

Nonfinite constructions are a prime example of both the unity and the diversity of the Germanic languages. Although they exhibit a wide range of language-internal variation, several core properties can be identified that remain remarkably stable and correlate with other characteristic features, particularly the basic word order within the VP (OV vs. VO). In this chapter, a descriptive overview of nonfinite constructions in the Germanic languages is provided. The focus is therefore on empirical generalizations that can be established within this domain rather than on analyses proposed in specific grammatical frameworks. This does not imply a wholly 'agnostic' perspective, however: standard concepts that are useful for a precise description of the relevant phenomena are employed, including notions such as (in)coherence and the distinction between raising and control.

Before outlining the structure of this chapter, let us begin with a definition: *Nonfinite constructions* (NFCs) are grammatical structures that contain at least one verbal predicate that is not marked for person or number, but may instead display specialized nonfinite morphology. This places infinitival, participial, and gerundial forms at the center of attention (see Sec. 2 for morphological aspects). From a typological perspective, it is astonishingly difficult to define finiteness vs. nonfiniteness (Nikolaeva 2007). Some of these problems may also extend to Germanic: Generally, there is a tendency for verbal person and number to erode while more relevant features like tense or mood (in the sense of Bybee 1985) may be preserved:[1] In the more conservative Germanic languages German (OV) and Icelandic (VO), person and

[1] According to Bybee, relevance constitutes a case of diagrammatic iconism. Categories that modify the deontation of a

Person/Number	**Icelandic**	**German**
1SG	*ég kasta*	*ich werfe*
2SG	*þú kastar*	*du wirfst*
3SG	*hún kastar*	*sie wirft*
1PL	*við köstum*	*wir werfen*
2PL	*þið kastið*	*ihr werft*
3PL	*þær kasta*	*sie werfen*

Table 1. Present forms of the verb ‘throw’ in Icelandic and German

number is robustly expressed, albeit with syncretism. This is illustrated with the verb ‘throw’ in Table 1 (for the 3SG, the feminine pronoun is used as a representative form, in Icelandic also in the plural).

By contrast, the other Germanic languages show different degrees of erosion: Excluding the highly suppletive verb ‘to be’, English only has a synthetic marker for the 3SG (*-s*); in Afrikaans, finite and non-finite forms are not formally distinguished (excluding the verbs *hê* ‘have’ and *wees* ‘be’) (Donaldson 1993: 218). Continental North Germanic shows a different, yet related situation in that no person/number marking can be found, yet infinitive and finite forms mostly do not coincide (cf. Vikner 2001: Ch. 1; Faarlund 2019: 6.1–6.2), as can be gathered from the Swedish examples in (1).

(1) *kasta* (INF), *kastar* (PRS), *kastade* (PST), *[har] kastat* (PTCP)

Whether tense is a semantic property of infinitival patterns is not so clear. Wurmbrand (2001) and Wurmbrand & Christopoulos (2020) argue that infinitives in Germanic languages do not encode tense as a syntactic category, but can still receive temporal interpretations depending on their structural size and embedding. Coherent (~ ‘restructuring’) infinitives lack independent temporal reference, while clausal infinitives can support richer, context-dependent temporal readings. Thus, apparent tense effects arise from structure, modality, and embedding rather than from an inherent tense feature in the infinitive itself. Future infinitives in German are robustly attested in corpora, although native speaker acceptance remains inconsistent (Reiner 2015). In (2), *spielen zu werden* with the nonfinite future auxiliary *werden* would be a bona-fide example of the future infinitive.

(2) Dem widersprachen die Spieler und betonten, auch ohne Geld für ihr Land spielen
this.DAT objected the players and stressed also without money for their country play
zu werden.
to become
“The players objected to this and stressed that they would play for their country even without remuneration.” (Nürnberger Zeitung, 16 June 2006; Reiner 2015: 508)

Rejecting a modal interpretation, Reiner (2015, 2024) claims that a future infinitive is emerging in present-day German as a temporal category to fill a paradigmatic gap for expressing posteriority (i.e. a genuine

verbal base to a higher extent need to be formally closer to the stem. In terms of exponence, this means that periphrastic expression or even absence is typical for lower-relevance categories whereas highly relevant categories can even show fusional expression, e.g. tense via ablaut or mood via umlaut.

temporal relation where E 'event time', coinciding with R 'reference time', is located after S 'speech time': $S < R = E$). However, she acknowledges that this semantic configuration is frequently categorized as prospective aspect in other theoretical frameworks.

Note that while the dependent(s) have to be nonfinite and verbal, this does not necessarily apply to the governing category. NFCs can be embedded under nouns (3), adjectives (4) and, albeit rarely, even prepositions (5).

(3) German (OV), Swedish (VO):

a. Die *Versuchung*, [an die Schnapsflasche zu gehen]
the temptation at the liquor-bottle to go

b. *Frestelsen* [att ta till spritflaska-n]
temptation=DEF IM take to liquor-bottle-DEF
"The temptation to reach for the bottle."

(4) a. Er ist *stolz* darauf, [sie zu kennen.]
He is proud CORR her to know

b. Han är *stolt* över [att känna henne.]
He is proud over IM know her
"He is proud to know her."

(5) a. Sie kritisierte das Buch, *ohne* [es gelesen zu haben.]
she criticized the book without it read.PTCP to have

b. Hon kritiserade bok-en *utan* [att ha läst den.]
she criticized book-DEF without IM have read.SUP it
"She criticized the book without reading it."

Let us now turn to a review of several diatheses in Germanic that are associated with NFCs. First, all Germanic languages permit passive constructions in nonfinite contexts, as illustrated by the examples from German (OV) and Swedish (VO). As (6b) demonstrates, the synthetic passive characteristic of the (Continental) North Germanic languages (see Harbert 2007: 327–329 for a brief overview) is likewise attested in these contexts.

(6) German, Swedish:

a. Das Buch verdient es, gelesen zu werden.
the book deserves it.CORR read.PTCP to become.PASS
"The book deserves to be read."

b. Bok-en förtjänar att läsa-s.
book-DEF deserves IM read-PASS

A particular subtype is the so-called modal passive, which is found predominantly in the West Germanic languages, as the examples in (7) show (see Demske-Neumann 1994 for a detailed discussion of the construction in German; Broekhuis, Corver & Vos 2015: 73 and Broekhuis & Corver 2015: 1009 on Dutch). This construction typically expresses 'root' modality, that is, deontic or dynamic interpretations.

(7) a. Das Buch ist zu lesen. (German)
"the book is to be read"

b. Het boek is te lezen. (Dutch)

c. Die boek is te lees. (Afrikaans)

A morphologically not-marked diathesis that seems to be found in most Germanic languages is the so-called causative passive, as the following examples from German (OV) and Danish (VO) in (8)–(9) illustrate. In English, this diathesis only seems possible with causative *have* that appears with the participle in this context.

(8) German (Gunkel 1999: 133–134):

a. Ich lasse Karl die Blumen gießen
I.NOM have/let Karl.ACC the flowers.ACC water.INF
"I have/let Karl water the flowers"

b. Ich lasse die Blumen (von Karl) gießen
I.NOM have the flowers.ACC water (by Karl)
"I have the flowers watered"

(9) Danish:

a. Han lod tømrer-en reparere dør-en.
he let carpenter-DEF repair door-DEF
"He has/let the carpenter repair the door."

b. Han lod dør-en reparere (af tømrer-en).
he let door-DEF repair (by carpenter-DEF)
"He has the door repaired (by the carpenter)."

A puzzling passive variant that is only observed in German is the so-called *Fernpassiv* (Höhle 1978: 176). It appears with optionally coherent matrix predicates like *versuchen*. As the contrast between (10a) und (10b) shows, passivizing the matrix verb can yield a passive conversion with the embedded predicate even though the accusative and an impersonal passive would be expected. It is regarded as a sign of cluster formation and unification of the thematic grids of the involved verbs. However, acceptability judgements for this construction seem to vary considerably (see the data reported by Wöllstein-Leisten 2001: 300–318).

(10) a. wenn Karl den Wagen zu reparieren versucht
if Karl the car.ACC to repair tries
"if Karl tries to repair the car"

b. wenn ? der/den Wagen zu reparieren versucht wird
if the.NOM/ACC car.NOM/ACC to repair try.PTCP become.PASS
"if the car is tried to be repaired"

The remainder of this chapter is structured as follows. First, the morphology of nonfinite forms is examined in greater detail (Sec. 2). We then turn to relational aspects (Sec. 3), including the notoriously difficult distinction between sentential (so-called 'incoherent') and non-sentential (so-called 'coherent') constructions, along with relevant diagnostics (Sec. 3.1), as well as thematic dependencies between matrix and embedded predicates, i.e. raising vs. control (Sec. 3.2). Next, we discuss non-sentential NFCs in more detail (Sec. 4), focusing on differences between OV and VO in this respect (Sec. 4.2), the 'verb cluster' property of Continental West Germanic (German, Dutch, Frisian), and morphosyntactic alternations such as the well-known IPP effect ('infinitive instead of (an expected) participle'), which can be observed in these types of complementation (Sec. 4.3). We then turn to formal and functional properties of sentential NFCs (Sec. 5), including their respective complementizers and the clause types and syntactic functions in which they appear (Secs. 5.1, 5.2). A brief summary and an overview of open questions conclude the chapter.

In terms of the languages consulted, we aim to provide sufficient coverage of both North and West Germanic. The following grammars are primarily consulted:

- German: Haider (2010), IDS-Grammar (Zifonun, Hoffmann, Strecker, et al. 1997)
- Dutch: Zwart (2011), Broekhuis & Corver (2015), Broekhuis, Corver & Vos (2015)
- Icelandic: Kress (1982), Thráinsson (2007)
- Afrikaans: Donaldson (1993)
- Continental North Germanic: Faarlund (2019), Holmes & Hinchliffe (2013), Lundskær-Nielsen & Holmes (2010)

2 The morphology of infinite forms in Germanic

Before turning to a closer examination of nonfinite patterns, it is necessary to consider the relevant morphological devices. A particularly useful framework for describing nonfinite patterns is Bech's theory of status and status government (Bech 1955). He distinguishes two *Stufen* ('levels'), namely *Supinum* 'supine' and *Partizipium* 'participle', each of which is associated with three distinct morphological *Status* ('states'), see Table 2.

	Level 1: supine	**Level 2: participle**
1st status	*lesen* 'read'	*lesend-*
2nd status	*zu lesen* 'to read'	*zu lesend-*
3rd status	*gelesen* 'read' (PTCP)	*gelesen-*

Table 2. Status theory (Bech 1955: 12)

With status government, Bech captures the fact that within complex (verbal) predicates, selectional restrictions between the different verbs hold. Let us have a look at a few German and Dutch examples: Modals, for instance, usually select for the 1st status, see (11), whereas the 3rd status is typical of the periphrastic perfect or past perfect, see (12). The 2nd status is typical of control constructions and certain other configurations, cf. (13).

(11) German, Dutch:

a. Der Lokführer muss *schlafen.*

b. De machinist moet *slapen*.INF
the locomotive.engineer must sleep
"The locomotive engineer must sleep"

(12) a. Der Lokführer hat *geschlafen*.

b. De machinist heeft *geslapen*.PTCP
the locomotive.engineer has slept.
"The locomotive engineer has slept."

(13) a. Der Lokführer versucht *zu schlafen.*

b. De machinist probeert *te slapen.*
the locomotive.engineer tries to sleep
"The locomotive engineer tries to sleep."

The most important point is that status government still holds when the governing verb is itself status-governed, as example (14) from German shows (adapted from Wöllstein-Leisten et al. 1997: 67): *hat* 'has' as a perfect auxiliary selects for a past participle of *behaupten* 'claim', which itself status-governs the *zu*-infinitive on *können* 'can'; the modal in turn governs the 1st status of *reparieren*. By means of status government, it can also be determined in which hierarchical order verbal elements stand. This order can be marked via indices, with the most deeply embedded verb always bearing the highest index (this is further discussed in Sec. 4.2).

(14) Der Lokführer hat alles reparieren zu können behauptet.
the locomotive.engineer has everything repair to could claimed
"The locomotive engineer claimed to be able to repair everything."

The distinction between the two levels (*Stufen*) is morphosyntactically motivated. The forms of level 2, although they bear striking similarities to those of level 1—most notably in their partially overlapping formal exponents—, are essentially adjectival in nature, as they typically function as nominal modifiers, cf. the German examples in (15). They are therefore related to the verbal forms of level 1 by conversion. Such forms will largely be disregarded in the following discussion.[2]

(15) German:

a. der am Kamin les-end-e Autor
the at=the fireplace read-PTCP-NOM.SG.M.WK author
"the author reading by the fireplace"

b. das bis Freitag zu les-end-e Buch
the till Friday to read-PTCP-NOM.SG.N.WK book
"the book to be read by Friday"

[2] In the traditional grammatical description of the North Germanic languages, uninflected participles, as they typically appear in compound tenses, are called *supines*, whereas the term 'participle' refers to their inflected counterparts in predicative (or stative-passive) use, e.g. Swedish *Jag har målat huset* 'I have painted the house' vs. *Det målade huset är till salu* 'The (newly) painted house is for sale'. The present participle has disappeared in most High German dialects (Weiß 2017).

c. die letztes Jahr geles-en-en Bücher
the last year read-PTCP-NOM.SG.N.WK books
"the books read last year"

2.1 Basic types of nonfinite forms

Even though Bech's morphological system was tailored to German, it is a useful frame for describing the situation in other Germanic languages as well. All Germanic languages exhibit a certain degree of overlap in terms of the nonfinite forms used in the different states:

- There is a simple infinitive (usually marked with an infinitival suffix);
- there are infintives marked with *zu* (and functional equivalents, see below);
- there are uninflected participles used for different periphrastic constructions, e.g. compound tenses like the perfect or the eventive passive.

This basic observation comes with several disclaimers, of course. Turning to the first status, the infinitival suffix can be altogether missing—this applies both to the first and (to a lesser extent) the second status. Typical examples are English or Afrikaans[3] (cf. Donaldson 1993: 217–218), but also certain High German dialects show this property (cf. the map in Rowley 2025: 81). In all North Germanic languages, *å/at(t)/að*, the cognate of the proximal Germanic preposition *at*, is used instead of the allative preposition *te/zu* common in West Germanic (Faarlund 2004), see (16a) vs. (16b)–(16c).

(16) a. *werk* 'work' – *te werk* 'to work' (Afrikaans)
b. *arbeta – att arbeta* (Swedish)
c. *vinna* 'work' – *að vinna* (Icelandic)

In both cases, these markers are the result of grammaticalization of the respective prepositions, a very common grammaticalization path (see Haspelmath 1989 and Demske 2001 on *zu* in German; Kuteva et al. 2019: 49, 351).

From a syntactic perspective, the second status marker is particularly interesting: In English and North Germanic, it clearly behaves like an independent syntactic element, whereas in Continental West Germanic it shows more resemblances to an affix or at least a special clitic (Schallert 2020; Cavirani-Pots 2020). There are several diagnostics for the different behavior of these markers: In English, two simple infinitives can be coordinated under *to* (17a), whereas this is highly restricted in German (17b) (examples from Haider 2010: 272). Additionally, *to* can be left behind in VP-ellipsis contexts, as (17c) shows (Haider 1993: 234). With particle verbs, i.e. verbs with separable first elements, *zu* can be positioned between particle and stem, very much like the participial prefix *ge-* (17d).

(17) a. He seemed to [$_{\text{VP}}$ laugh and cry] at the same time
b. Er schien gleichzeitig [zu lachen und *(zu) weinen]

[3] In Afrikaans, in most cases there is also a large overlap between the infinitive and most finite forms due to erosion of the respective bounded morphological markers.

c. They are [$_{VP}$ laying eggs now], just like they used to [$_{VP}$ ∅]

d. an=ge=fangen ‘started’, auf=zu=hören ‘to finish’

Most famously, English (*to boldly go…*) and North Germanic, see (18), allow split infinitives, i.e. non-verbal interveners, whereas in Continental West Germanic, this is highly restricted for *te/zu* in that this element can only be disconnected within the verbal complex or with incorporated nouns, as the examples from Zurich German in (19a) and Northern Dutch varieties in (19b) show (see Schallert 2020 on German dialects and Corver, Vos & Broekhuis 2015: 70 on Dutch).

(18) Icelandic, Norwegian [Nynorsk]:

a. að ekki lesa þessa bók strax.
to not read this book immediately
“to not read this book immediately” (K. R. Christensen 2007: 154)

b. Det er viktig å alltid komma i tide.
it is important to always come in time
“it is important to always come in time” (Emonds & Faarlund 2014: 98)

(19) Zurich German, Dutch:

a. Schämsch ∅ di nüüd cho z bättle
shame *pro* REFL not come to beg
“Aren’t you ashamed to come begging?” (Weber 1987: 244)

b. Marie zit te piano spelen.
Marie sits to piano play
“Marie is playing the piano” (Corver, Vos & Broekhuis 2015: 70)

A wider array of Continental West Germanic varieties retains a so-called ‘gerund’ (not to be confused with its English counterpart), i.e. a specialized infinitival suffix typically connected with the *zu*-infinitive, which can also appear in different other syntactic contexts (see Hoekstra 1997 on Frisian and Landolt 2025 for a recent overview on German dialects). Historically, these morphs come from inflected forms of the infinitive, yet there are also certain interences with present participle forms that largely ceased to exist independently in the dialects (Becker & Schallert 2021).

In the third state, there is a certain amount of variability: With the notable exception of Afrikaans, all Germanic languages exhibit two suffix allomorphs that are connected to inflectional classes: Regular (i.e. ‘weak’) verbs select a dental suffix (*-t*, *-ed*, etc.), while irregular (i.e. ‘strong’) verbs have *-en* (or comparable variants); (20a) features relevant examples of both cases.[4] In English and the Scandinavian languages, the prefix *ge-* was lost while in Continental West Germanic, there is a North-South division: Frisian and most parts of Low German abandoned the prefix, while it is retained in Dutch and German.[5] In Standard German, the appearance of *ge-* is phonologically conditioned in that it can only appear with

[4] I am ignoring different verbal subclasses, e.g. modals, that may show differences in other paradigm cells but pattern with one of the larger class forms in the participle.

[5] See e.g. map 132 of the *Sprachatlas des deutschen Reichs* [‘Linguistic Atlas of the German Empire’] on the participial form *gesagt* ‘said’: http://www.regionalsprache.de/Map/1N6aBRiG (accessed on 31 July 2026). Note that many Upper German dialects show schwa-syncope and ensuing total assimilation of the prefix in certain contexts, e.g. Bavarian *gleesn* ‘read’ (PTCP) vs. *broodn* ‘fried’ (PTCP).

monosyllabic and trochaic disyllabic stems (Wiese 2000: 92–93), as evidenced by the contrast between (20a, b) and (20c).

(20) a. *ge-mach-t* ‘made’ (PTCP) – *ge-worf-en* ‘thrown’ (PTCP)
b. *arbeit-en* ‘work’ [ˈʔaʁ.bai̯.tn̩] (INF) – *ge-arbeit-et* (PTCP)
c. *krakeelen* [kʁa.ˈkeːln̩] ‘brawl’ – *krakeelt* (PTCP)

In particular, prefix verbs never allow *ge-* (21a), while with particle verbs (which are separable and the particle bears secondary stress), the prefix is always sandwiched in between (21b). In Dutch, the distribution is roughly the same, yet certain non-trochaic stems also allow prefixation, e.g. *gefeliciteerd* ‘congratulated’ (Rathert 2009: 164).

(21) a. *verkaufen* [fɐ.ˈkaʊfn̩] ‘sell’ (INF) – * *ge-verkauft*, * *ver-ge-kauft* (PTCP)
b. *abkaufen* [ˈʔab̥.ˌkaʊfn̩] – *ab-ge-kauft*

Historically, *ge-* derived from an aspectual derivational marker that came to be grammaticalized as an inflectional prefix (see e.g. Wischer & Habermann 2004). In certain Central and East Upper German dialects there are also *ge*-prefixed infinitives which stem from the same historical source (cf. Paul et al. 2007: 267 [§ M 94, Rem. 2]); they mainly appear as complements to modal verbs (see e.g. Birkenes & Fleischer 2019: 459–460; Rowley 2025: 91–96). A relevant example is given in (22).

(22) East Franconian dialects (High German):
dǫs kho i nid gədrǭxə
that can I not ge=carry
“I can’t carry that.” (Rowley 2025: 93)

2.2 Further forms and distinctions

Apart from these basic contrasts there are several other nonfinite forms in Germanic stemming from different historical sources. For the sake of illustration, I pick three examples: the English gerund, the progressive markers *am* (German) and *aan het* (Dutch), and verb doubling in Alemannic dialects.

Let us start with one of the many oddities of English among its Germanic relatives: It has developed the so-called gerund construction, e.g. *Leave me where I am, I’m only sleeping* (Beatles). Historically, it is related to deverbal abstractive derivations that came to be re-connected to the verbal system (Fonteyn 2019, and many others). This phenomenon is the locus classicus when it comes to categorial ‘squishes’ due to its chameleon-like properties. While uses like (23a) are clearly nominal, as evidenced by the plural inflection, complement-taking ones like (23b) clearly belong to the verbal spectrum. However, there are also fuzzy cases as (23c) that combine a possessive element typical of NPs with a VP-typical nominal complement (examples from Quirk et al. 1985: 1290–1291; discussion after Aarts 2004: 17–18).

(23) a. some paintings of Brown’s
b. Brown is painting his daughter.

c. Brown's deftly painting his daughter is a delight to watch.

An important overlap with other infinitival constructions are control configurations like (24) where there is coreference between a nominal antecedent and an implicit referent in the gerundial construction (see Sec. 3.2 on control). In terms of its functional spectrum, gerunds show a certain overlap to 'absolute' participial constructions (i.e. non-attributive uses) in other Germanic languages, cf. the German example in (24b), but those patterns are largely restricted to Standard languages and/or rather formal registers (see e.g. Zifonun, Hoffmann, Strecker, et al. 1997: 2214–2230 for a detailed survey on these patterns in German).

(24) a. [PRO^{i} Having eaten a large bowl of pasta], John^{i} was unable to move for several hours.

b. Sie^{i} fotografierte ihn^{j}, [PRO^{i,j} die Straße überquerend.]
"She took a photo of him crossing the street"

The second example are the infinitival particles *am* in German and *aan het* (colloquially, *aan 't*) in Dutch for signaling imperfective (progressive) aspect (van Pottelberge 2004), see (25). These markers were grammaticalized from the positional-stative preposition *an/aan* and a clitic form of the definite article plus a nominalized form of the infinitive.[6]

(25) German, Dutch:

a. er ist am arbeiten
he is at=the work.INF
"He's working"

b. zij is aan het werken.INF
she is at the work
"She is working"

In regional and dialectal variants of German there is clear evidence that this construction is becoming further integrated into the verbal system, as evidenced by the fact that it shows a much larger distribution area with intransitive verbs (e.g. *schlafen* 'sleep') than with transitive verbs and excorporation after the model of (26b) (e.g. *reparieren* 'repair').[7] Clear signs of the verbal status of the infinitive is the fact that adjectival modifiers are unacceptable (26a) (examples from Ramelli 2015: 69, 132).

(26) a. * Er ist am lauten Singen.
he is at=the loud singing

b. und dann war der sich da für fünf Pfennig die Schnäpschen am hineinschütten
and then was he REFL there for five pennies the shots at=the knock.back
"And then he was knocking back little shots in there for five pfennigs."

Much attention was devoted to assessing with which aspectual classes of verbs this construction is attested, in particular whether it is compatible with telic predicates (e.g. Flick & Kuhmichel 2013), yet an

[6] In Afrikaans, there is a very similar construction featuring *wees* 'be' *aan* + *die* + verb, e.g. *Ek is 'n brief aan die skryf* 'I'm writing a letter' (Donaldson 1993: 221–222).

[7] See the respective maps on these two verbs in the *Atlas zur deutschen Alltagssprache* ['Linguistic Atlas of Colloquial German']: https://www.atlas-alltagssprache.de/runde-2/f18a-b/ (accessed on 31 July 2026).

important desideratum is the question how obligatory this aspectual construction is, i.e. whether it can always be substituted with a simple finite form ('paradigmatic variability' in the sense of Lehmann 2015: Ch. 4.2.3).

The third phenomenon I want to mention in this respect is the so-called 'verb-doubling' construction that can be found in many Alemannic dialects, most commonly with the verbs *choo* 'come' and *goo* 'go' (Lötscher 1993), as evidenced by (27a). Note also that there is a quite similar construction in West Flemish, yet only with the motion verb *goan* 'go' (Haegeman 1990). Due to their large degree of variability, the synchronic status of such constructions is hard to assess: While some cases can be described as syntactic reduplication, in others the apparent duplifix clearly stems from a prepositional element that is still independently existent, as the parallelism between (27b) and (27c) shows (see Schallert 2010: 54–57; Brandner & Salzmann 2012).

(27) Swiss Alemannic, Vorarlberg Alemannic:

a. Er chunnt cho der Onkel bsueche.
He comes comes the uncle visit
'He comes to visit his uncle.' (Lötscher 1993: 180)

b. es kunnt gi regna
EXPL comes GO rain
'It's about to rain.'

c. Sie fahrt gi Feldkirch
she drives GO Feldkirch
'She's driving to Feldkirch'

Lastly, in certain syntactic contexts, morphological alternations and/or specialized forms can appear. Two well-known cases are the so-called IPP-effect (infinitivus pro participio, i.e. 'substitute infinitive') and its mirror-image, the PPI-effect (participium pro infinitive, i.e. 'substitute participle'). I will briefly address these and related phenomena in Sec. 4.3. Chiefly in West Germanic, we also find specialized infinitival complementizers (e.g. *um/om* in German/Dutch), which are discussed in Sec. 5.

3 Relational aspects

In this section, we review two central dimensions in the typology of nonfinite constructions. The first concerns *coherence*, that is, whether a construction exhibits a sentential (= incoherent) or a nonsentential (= coherent) configuration. All Germanic languages display nonfinite constructions that can unequivocally be classified as sentential, as evidenced e.g. by the presence of dedicated complementizers or structural evidence for a fixed subject position (cf. Haider 2010: 302). However, the Germanic OV languages (e.g. Dutch and German) differ from their VO counterparts in the so-called cluster property (Haider 2003; see also Sec. 4): in coherent constructions, complex verbal predicates may form a compact verbal cluster in certain configurations, whereas in incoherent constructions nonverbal material may intervene between the verbal elements.

The second dimension concerns the argument structure of the predicates involved and the dependencies between them. In control constructions, the theta-grids of the matrix and embedded predicates are satisfied independently, although coreference relations may obtain between the matrix subject or other grammatical functions and the typically implicit subject of the embedded clause. In raising constructions, by contrast, the matrix predicate does not assign a theta-role to its subject. Instead, the matrix subject originates as the external argument of the embedded predicate and is raised into the matrix subject position. As a cover term for raising/control, we use Bech's (1955) notion of *orientation*. We discuss some diagnostics for these four categories (i.e. ± coherence, ± control) and highlight their typological significance for the Germanic languages.

3.1 Clausal status: incoherence vs. coherence

A central question around nonfinite complementation is the following: Are we dealing with a sentential (incoherent) or non-sentential (coherent) configuration? Bech (1955) coined the terms 'coherence' and 'incoherence' for this state of affairs, with special reference to German, and he also developed different diagnostics for the two construction types (see von Stechow 1984; Schallert 2014b: Ch. 2.1 for further discussion). We do not further address his original definitions but cut straight to the chase.

In the Germanic VO languages, infinitival complementation can involve any of the basic complementation types, viz. VP (28a), TP (28b), and CP (28c) (Haider 2010: 298, 308; examples partially taken from him).

(28) a. He has [$_{\text{VP}}$ been [$_{\text{VP}}$ criticized by nearly everyone]]
b. She seemed/appeared [$_{\text{TP}}$ to [$_{\text{VP}}$ understand the problem]]
c. She threatened [$_{\text{CP}}$ PRO to play the violin piece once more].

In the Germanic OV languages, by contrast, there is a main division between the cluster property, where the verbs in the right periphery form a compact unit, on the one hand and sentential embeddings on the other hand (CP). The empirical situation in German/Dutch is further discussed in Sec. 4.1. At this point, we focus on basic diagnostics that can be used for the OV languages (Müller 2002: Ch. 2.1.2; Haider 2010: 311–321). We restrict attention to the following three tests, with a final remark on a semantic criterion (scope of adjuncts) at the end of this section:

1. Extraposition
2. Scrambling
3. Topicalization

Let us first note that incoherence only applies for a subset of verbs that select a *zu*-infinitive (first noted by Bech 1955: 68 in the context of his *Kohärenzregel* 'coherence rule'), meaning that predicates that select for other nonfinite forms (in particular a bare infinitive or a participle) always construe coherently.[8] The classical test for incoherence is extraposition: If the infinitival structure can appear in the position

[8] This means that with verbs like *(sich) trauen* 'dare' or *helfen* 'help' with optional *zu*-marking, the variant with normal infinitive (1st status) is coherent (see Reis & Sternefeld 2004: 498). German dialects like Alemannic show null infinitives with a range of matrix predicates while also allowing the third construction (see Schallert 2013 for an overview). Selection of *zu* is a necessary, yet not a sufficient criterion because there are several verbs in this class which construe (obligatorily) coherently, e.g. raising predicates like *scheinen* 'seem', control predicates like *wissen* 'know' or semi-modal usages of *drohen* 'threaten' or *versprechen* 'promise' (Reis & Sternefeld 2004: 471, 474 on German; Zwart 2011: 365 on Dutch).

after the right sentence bracket (RSB), which typically hosts the lexical verb in compound predicate forms, it is incoherent, see the contrast between (29a) and (29b). More specifically, extraposition is traditionally considered as evidence for CP- (and fully clausal) status. Conversely, coherent NFCs are typically intraposed (30a), vs. (30b).

(29) a. Karl hat [das Buch zu lesen] versucht.
Karl has the book to read tried

b. Karl hat *versucht*, [das Buch zu lesen].
Karl has tried the book to read
"Karl tried to read the book."

(30) a. weil Karl [das Buch zu lesen] scheint
because Karl the book to read seems

b. * weil Karl scheint [das Buch zu lesen]
because Karl seems the book to read
"because Karl seems to read the book"

This test is faced with certain problems, however. In German, incoherent structures can also appear in the middle field, cf. (31a). Note that in this case, cluster formation cannot have taken place because the verbal complex is interrupted by the (sentential) negation marker *nicht*. In Dutch, by contrast, full infinitival clauses must be extraposed, see (31b), or can appear in the prefield in some cases (Haider 2010: 283; Zwart 2011: 110–112).

(31) a. Hoffentlich hat sie [ihn zu informieren] nicht vergessen.
hopefully has she him.ACC to inform not forgotten
"Hopefully, she has not forgotten to inform him."

b. Tasman heeft geprobeerd (om) het Zuidland te vind-en.
Tasman has tried.PTCP COMP the south.land to find
" Tasman tried to find the South Land." (after Zwart 2011: 110)

Scrambling: This term refers to order variability in the Mittelfeld 'midfield' (see e.g. Haider & Rosengren 2003). In German, this phenomenon is strictly clause-bound. Therefore, coherent constructions allow for the objects of the infinitive to behave as clause-mates of the matrix verb and can freely permute with matrix arguments: In (32a), for instance, the direct object *es* 'it' appears before the matrix subject *jemand* 'somebody' and the matrix indirect object *ihm* 'him'. The same applies to the direct object *den Aufsatz* (a full NP) with regard to the matrix subject *jemand* 'somebody' in (32b).

(32) a. weil es ihm jemand zu lesen versprochen hat
because it.ACC him.DAT somebody to read promised has
"because somebody promised him to read it"

b. weil ihm den Aufsatz jemand zu lesen versprochen hat
because him.ACC the.ACC Aufsatz.ACC somebody.NOM to read promised has
"because somebody promised him to read the paper"

Topicalization: This phenomenon involves moving a constituent to the position of the finite verb in main clauses ('Vorfeld' in topological terms). Topicalizing a complete infinitival phrase (the entire VP or CP including its objects) as a single unit indicates that it forms a clausal (incoherent) structure, see (33a) (note that this example is perfectly grammatical even though it is information-structurally marked). However, the verb of the infinitival clause cannot be extracted alone (33b). Conversely, fronting non-verbal constituents (or lower portions of the verbal complex) is licit with the coherent construction, see (33c, d).

(33) a. [$_{CP}$ Ihr etwas mitzuteilen versucht] hat er.
her something with=to=tell tried has he
"He tried to tell her something."

b. * Mitzuteilen$_i$ hat er versucht [$_{CP}$ ihr etwas e$_i$]

c. Ihr$_i$ hat er versucht [$_{CP}$ e$_i$ etwas mitzuteilen]

d. [Mitzuteilen]$_i$ hat er ihr etwas versucht e$_i$

An intriguing phenomenon in Continental West Germanic is the so-called *third construction* (a term coined by den Besten & Rutten 1989). In this construction type it looks as if material from the embedded infinitival clause ends up in the matrix clause. It can be observed with a subset of control verbs that also allow for the incoherent construction and even certain raising predicates (Wöllstein-Leisten 2001; Reis 2007; Zwart 2011: 365–366 on the relevant predicates in Dutch). However, there are clear signs that this construction patterns with the coherent type, as can be seen by e.g. scrambling or pronoun fronting (Reis & Sternefeld 2004: 472):

(34) weil ihm | dem Jungen KEIner versucht ein Auto zu verkaufen
because him.DAT the boy.DAT nobody triers a car to sell
"because nobody tried to sell him/the boy a car."

Let us finally give an overview of the relevant classes of predicates in terms of their coherence properties.[9] We focus on German but in many respects, the situation in Dutch is comparable. We get three basic classes of predicates (classification drawn from Reis & Sternefeld 2004; Haider 2010 and Zwart 2011).

- *Obligatorily coherent (~ clustering) verbs* (Haider 2010: 275, 308; Zwart 2011: 363–365): Temporal and aspectual auxiliaries like *haben* 'have', *sein* 'be', *werden* 'become'; modal verbs, e.g. *müssen* 'must', *können* 'can' (see Reis 2001 for a detailed survey of their coherence properties). Note that auxiliaries are typically highly polysemous, e.g. *werden* plus infinitive (1st status) is used for the compound future or evidential-modal meanings, with the past participle (3rd status) it functions as passive auxiliary. Another important class are EMC verbs (see also Sec. 3.2), i.e. causatives like *lassen* or *machen* 'make' (archaic) or perception verbs like *sehen* 'see', *hören* 'hear'.
- *Obligatorily incoherent verbs:* This class is difficult to define because existing work has rather focused on syntactic criteria. Due to obligatory extraposition, for instance, a predicate like *geeignet sein* 'be apt/liable' can be classified as incoherent, cf. (35a) vs. (35b). Rapp & Wöllstein (2009: 160–161) see (syntactic) incoherence, in particular the presence of a C-head, as

[9] In the following, I focus on verbal predicates. Note, however, that there are also nonverbal or complex predicates that select for nonfinite complements (see Sec. 1 for discussion).

precondition for referential anchoring of a clause, either via its propositional content or the presence of a presupposition, as is the case, for instance, with factive predicates like *leugnen* 'deny' or *bedauern* 'regret'. Zwart (2011: 366), who also takes obligatory extraposition as the relevant criterion, lists three relevant subgroups, i.e. predicates like *begrijpen* 'understand' without the complementizer *om*, predicates like *lukken* 'succeed' with *om* and a correlative pronoun (*het* 'it') in the matrix clause, and predicates like *aansporen* 'urge' with the complementizer, yet no correlative pronoun.

(35) a. Diese Aussage ist geeignet, seine Karriere zu zerstören.
this statement is apt/liable his career to destroy
"This statement is liable to destroy his career."

b. * Diese Aussage ist seine Karriere zu zerstören geeignet.

- *Optionally clustering verbs:* A bigger class of predicates selecting a *zu/te*-infinitive both allows for the incoherent as well as the coherent construction. This can be seen, for instance, with the verb *beabsichtigen* 'intend' in German (Haider 2010: 278–279): While (36a) can either be analyzed as an instance of the incoherent or the coherent construction, cf. the structural representations in (36b, c), the variants in (37a) can only be coherent due to partial fronting, while and (37b) is incoherent because the *zu*-infinitive and the rest of the verbal complex is interrupted by the adverbial *wirklich* 'really'. Relevant German examples for optionally coherent predicates are cited in Haider (2010: 279); for Dutch, the predicates mentioned by Zwart (2011: 365–366) that optionally allow for the third construction can be regarded as representative.

(36) a. dass er niemanden zu stören beabsichtigt hat
that he nobody.ACC to disturb intended has
"that he intended to disturb nobody"

b. dass er [$_{CP}$ niemanden zu stören] beabsichtigt hat

c. dass er niemanden [$_{VC}$ zu stören beabsichtigt hat]

(37) a. [$_{VC}$ Zu stören beabsichtigt] hat er wirklich niemanden
to disturb intended has he really nobody

b. dass er [$_{CP}$ niemanden zu stören] *wirklich* beabsichtigt hat

A final remark: As discussed, syntactic (in particular: topological) criteria have figured prominently in the demarcation between coherent and incoherent structures. However, there is also a longer debate in which respect semantic or pragmatic factors contribute to 'sentencehood'. A criterion for (in-)coherence located at the syntax-semantics interface is the scope of adjuncts and sentence negation (see e.g. Müller 2002: 39–40; Haider 2010: 315). While the (optionally) coherent structure in (38) allows for the negation to scope only over the embedded predicate (38a), also a reading with matrix scope is possible (38b). By contrast, (39) features complete extraposition of the infinitival complement and thus incoherence. In this case, the scopal domain of the negation is the respective clause.

(38) weil wir den verrückten Wissenschaftler nicht zu stören versuchten
because we the mad scientist not to disturb tried

a. *versuchen*(¬(*stören*))

b. ¬(*versuchen*(*stören*))

(39) weil wir (*nicht*) versuchten, den verrückten Wissenschaftler (*nicht*) zu stören

Stiebels (2010: 414) regards *event coherence*, i.e. "the integration of two sub-events into a coherent overall event" (English translation by O. S.) as an important criterion for semantic (des-)integration. This property can be tested, for example, by examining the extent to which the verb embedded by the matrix predicate allows for its own temporal reference or not. This criterion identifies phase predicates following the pattern of *anfangen* 'begin' in (40) as event-coherent, since they isolate a sub-event denoted by the embedded predicate and thus cannot refer to a distinct sub-event (see Stiebels 2010: 421).

(40) a. # weil er gestern anfing, heute die abgefallenen Nüsse aufzusammeln
because he yesterday began tomorrow the fallen nuts to collect

b. # weil sie in der Küche anfing, im Garten ein Buch zu lesen
because she in the kitchen began in=the garden a book to read
(Stiebels 2010: 421)

The situation is different with commissive or directive predicates such as *versprechen* 'promise' or *bitten* 'ask' because, as the examples (41) show, independent local and temporal modification is possible here without any problems.

(41) a. Sie hat mir gestern versprochen, heute das neue Betriebssystem zu installieren.
she has me.DAT yesterday promised today the new operating.system to install
"She promised me yesterday to install the new operating system today."

b. Sie hat mir im Zoo versprochen, zuhause einen Kuchen zu backen.
she has me.DAT in=the zoo promised at=home a cake to bake
"She promised me in the zoo to bake a cake at home." (Stiebels 2010: 423)

Various studies on infinitive syntax suggest that event coherence also requires a closer syntactic connection between the predicates involved (e.g. a subclass of so-called 'exceptional case-marking' [ECM] predicates; see Sec. 3.2), while a distinct temporal reference of the embedded predicate is regarded as an indication of syntactic disintegration. Wurmbrand (2001), for example, assumes that the structural differences between infinitival classes—specifically VP, vP/TP, and CP—correspond to semantic interpretations such as irrealis, factive, and propositional. She argues that these semantic types determine the distribution of restructuring, with specific factors like the factivity presupposition necessitating a full CP structure.

From a typological perspective, Cristofaro (2003) posits a *Complement Deranking Hierarchy*, see (42). It is essentially semantically motivated and, in addition to semantic integration, also makes statements about the specific verb form of the complement. The higher the corresponding predicate class is in the hierarchy, the more likely it is to use verb forms that do not occur in independent clauses—characterized, for example, by the absence of morphological distinctions, particularly finiteness markers (see Cristofaro 2003: 55).

(42) Complement Deranking Hierarchy (Cristofaro 2003: 125):
Modals, Phasals > Desideratives, Manipulatives (‘make’, ‘order’) > Perception > Knowledge, Propositional attitude, Utterance

3.2 Orientation: raising vs. control

The second key dimension for the analysis of nonfinite constructions (NFCs) is the distinction between raising and control, which concerns the thematic relationship between the matrix predicate and the embedded predicate. In control constructions, the argument structures of the matrix and embedded predicates are semantically independent. Consequently, each predicate assigns its own theta roles, including an external argument. At the same time, one of the matrix arguments is obligatorily interpreted as coreferential with the implicit external argument of the embedded clause. As illustrated in (43a), *Oliver* is the external argument of the matrix verb *try* (A_1), while the embedded predicate *understand* also requires its own external argument (E_1). Because nonfinite clauses cannot realize an overt subject, this argument remains unexpressed; following the Government and Binding tradition, it is represented as PRO (see below). The obligatory identity relation between A_1 and E_1, indicated via a superscripted index in (43b), is the defining property of control constructions; propositional arguments are semantically labeled as SOAs (‘state of affairs’).

(43) a. [$_{\theta}$ Oliver] tries [$_{CP}$ [$_{\theta}$ PRO] to understand nonfinite constructions.]
b. *try*($A_1{}^i$, A_2:SOA), *understand:*SOA($E_1{}^i$, E_2)

There are three basic interpretations for control relations: subject (44a), object (44b), and arbitrary control (44c). Subject control means that the matrix subject is interpreted as coreferent with the (non-overt, implicit) subject of the infinitive, with object control it is the object—depending on the case system of the respective language, this can also be the indirect object (45a) or prepositional objects (45b). Arbitrary control refers to configurations where no suitable controller is available, indicated in (44c) with PRO^{arb}.

(44) a. The abbot^{i} tried [PRO^{i} to solve the problem]
b. The abbot advised the priest^{i} [PRO^{i} to solve the problem]
c. [PRO^{arb} To be patient] is important.

(45) a. Der Abt hat ihm^{i} geraten, [PRO^{i} sich ruhig zu verhalten.]
the abbot has him.DAT advised REFL calm to behave
“The abbot advised him to keep a low profile.”
b. Der Abt hat [$_{PP}$ auf ihn]^{i} eingeredet, [PRO^{i} sich ruhig zu verhalten.]
the abbot has on him talk.into REFL calm to behave
“The abbot persuaded him to stay calm.”

A theory-neutral definition of control that encompasses different other interpretations can be found in Stiebels (2007: 13), as given below.[10] In this representation, the arguments *X* and *Y* represent the con-

[10] Note that the notion of control can also be transferred to certain finite embeddings. See Landau (2004) and Stiebels (2010) for some further discussion.

troller, and Z the controlled argument; the various associated control interpretations can be specified by (co-) indexing the respective arguments.

> OC applies to structures in which a predicate P_1 selects an SOA-argument and requires one of its (individual) arguments to be (improperly) included in the set of referents of an argument of the embedded predicate P_2 heading the SOA-argument.
> $[X_i\ P_1\ (Y_j)\ [_{\text{SOA}}\ Z_k\ P_2\ \ldots]]$ with $k \cap \{i,j\} \neq \emptyset$

Thus, exhaustive control, as in (44a) above, requires referential identity between the controller and the controlled argument ($k = i$, $k = j$). In partial control, following the pattern of (46a), however, the controlled argument includes, in addition to the controller, further referents ($k = i + v$ or $k = j + v$), whereas in split control (46b), it is controlled jointly by two argument referents (cf. Stiebels 2007: 5–6).

(46) a. John^{i} wanted [PRO^{i+v} to meet at six.]
b. Maria^{i} agreed with Peter^{j} [PRO^{i+j} to publish the book together.]

Since nonfinite constructions typically lack an overt subject argument, it is represented as a gap or—in traditional GB terms—as the empty category PRO, which is subject to both principles A and B of the Binding theory, meaning that it both has anaphoric and pronominal properties (Chomsky 1981: 60, 64): Obligatory control means it needs a controller in its governing category, as is typical with reflexives. On the other hand it also has pronominal traits in that it must not be bound in its binding domain.

There are different arguments for representing the missing subject of control constructions as an empty PRO-form. In German, for instance, appositions like *einer nach dem anderen* ‘one after the other’ have to match the case of their target expression (Höhle 2019: Sec. 6; see also Müller 2002: 49–53). In (47a) this is illustrated with the pluralic subject *Die Männer* ‘the men’ where the nominative is copied; (47b) shows the same relation with the accusative.[11]

(47) a. Die Männer stiegen einer nach dem anderen aus.
the men got one.NOM after the other out
“The men got out one by one”
b. Wir begrüßten die Männer einen nach dem anderen.
we welcomed the men one.ACC after the other
‘We greeted the men one by one.’

With an uncontroversially incoherent infinitival construction like (48a), the only overt antecedent would be *den Männern* ‘the men’ (DAT), yet the apposition appears in the nominative, while the dative is ungrammatical (48b).

(48) Ich habe den Männern geraten, im Abstand von ein paar Wochen, …
I have the men.DAT recommended in=the distance of a few weeks
a. einer nach dem anderen zu kündigen.
one after the other to resign

[11] Note that this argument is only valid if it is assumed that PRO is case-marked, which was explicitly ruled out in Chomsky (1981: 181).

b. * einem nach dem anderen zu kündigen
one.DAT after the other to resign
"I advised the men to quit one by one, a few weeks apart."

Raising configurations, on the other hand, involve a different thematic relation between matrix and embedded predicate. Subject raising predicates like *seem* do not subcategorize for a subject argument but instead 'copy' or attract the highest argument of the embedded predicate. This is illustrated in (49a, b) where the experiencer of *love* appears in the subject position of the matrix clause.

(49) a. [$_\theta$ Oliver$_i$] seems [[t$_i$] to love model trains.]
b. *seem*(E_1, A_1:SOA), *love:*SOA(E_1:$\emptyset$, E_2)

A typical diagnostic of raising involves embedded predicates with expletive subjects, lacking a corresponding semantic argument, as evidenced by (50a): The expletive *it* belongs to the embedded, 0-valued predicate *rain*. Note that with finite embeddings like (50b), the expletive is associated with the matrix predicate *seem*. A related test involves idiom chunks, which are licensed by raising predicates (51a) but ungrammatical with control predicates (51b).

(50) a. It seems to have rained.
b. It seems [that Oliver likes model trains].

(51) a. The shit seems to hit the fan.
b. # The shit tries to hit the fan.

Apart from subject raising predicates there is also another class, i.e. raising to object predicates. Typical cases involve causatives (52a) and perception verbs (52b). While modern analyses in different grammatical frameworks may differ in several aspects, the basic intuition is that also in these cases, arguments of the matrix and embedded predicate are shared (or: 'raised' to the object position of the matrix predicate).[12] In traditional Government-and-Binding terminology, raising to object is called ECM ('exceptional case marking'), typically with causatives (52) or perception verbs (52), because the highest argument of the embedded predicate appears in the objective case. Another possibility for subject licensing observed in English is the prepositional complementizer *for*, as evidenced in (52c) with an experiencer argument; see also the discussion in Sec. 5.

(52) a. We let [John paint the door.]
b. We heard [John whistle a nice tune.]
c. I want [[for John] to be happy.]

In the following, we introduce a simplified version of the HPSG formalism (HPSG light, cf. Müller 2023) for representing and analyzing raising and control configurations; then, we discuss some basic

[12] The first generative analysis of different raising constructions in found in Postal (1974). Bech (1955) does not distinguish between raising and control, even though he was fully aware of the relevant facts. What have come to be regarded as modern notions are subsumed under his flexible concept of Orientierung 'orientation' (see von Stechow 1984: Sec. IV for a very enlightening and concise discussion).

properties of both construction types in Germanic. In HPSG-light, *raising* and *control* constructions are distinguished by the relation between syntactic argument structure and semantic role assignment. In a raising construction such as *Oliver seems to love model trains*, the matrix verb does not assign a semantic role to its subject; instead, the subject is inherited from the embedded predicate via structure sharing on the SUBJ list:

(53) *seem* ⇒ $\begin{bmatrix} \textsc{subj} & \langle \boxed{1} \rangle \\ \textsc{comps} & \left\langle \left[\textsc{subj} \ \langle \boxed{1} \rangle \right] \right\rangle \end{bmatrix}$

The NP *Oliver* therefore receives its semantic role only from the embedded verb *like*. By contrast, in a control construction such as *Oliver tries to understand nonfinite constructions*, the matrix verb assigns its own semantic role to the subject while simultaneously identifying it with the understood subject of the infinitival complement:

(54) *try* ⇒ $\begin{bmatrix} \textsc{subj} & \langle \boxed{1} \rangle \\ \textsc{comps} & \left\langle \left[\textsc{subj} \ \langle \boxed{1} \rangle \right] \right\rangle \\ \textsc{cont} & \textit{try-rel}\left(\boxed{1}, \ldots \right) \end{bmatrix}$

The crucial difference is therefore that raising involves only syntactic argument sharing, whereas control additionally involves semantic selection by the matrix predicate.

Let us first observe the spectrum of control predicates in Germanic. The classification by Sag & Pollard (1991: 65–66, 78), which is based on Comrie (1984), distinguishes between three basic types:

1. Influence predicates (*persuade*, *appeal*): [INFLUENCE, INFLUENCED^{i}, SOA-A^{i}]
2. Commissive predicates (*promise*, *try*): [COMMITOR^{i}, SOA-A^{i}]
3. Orientation predicates (*want*, *hate*): [EXPERIENCER^{i}, SOA-A^{i}]

Influence verbs mark directive speech acts and usually involve object control; commissive predicates, by contrast, typically feature subject control. With orientation verbs, the controller corresponds to the experiencer of the matrix predicate. In all Germanic languages, we find lexical correspondences to the English examples quoted above, e.g. *vragen* (Dutch) 'ask', *neyða* 'force' (Icelandic) for group 1; *versprechen* 'promise' (German), *prøve* 'try' (Norwegian [Bokmål]) for group 2; and *haat* 'hate' (Afrikaans), *wel* 'want' (North Frisian [Fering]) for group 3.

Regarding ECM-predicates, we can distinguish two basic types, which we call ECM_1 and ECM_2, respectively (Wurmbrand & Christopoulos 2020). The first class, ECM_1, always occurs without an infinitive marker, and the matrix verb selects a tenseless eventuality (55a). The second class, ECM_2, selects full propositions, as evidenced, for example, by the possibility of independent temporal reference in the embedded clause (55b).

(55) a. I saw/let him steal ice cream. (Wurmbrand & Christopoulos 2020: 393)
b. The bridge is expected to collapse (tomorrow). (Wurmbrand & Christopoulos 2020: 391)

ECM_1-verbs (mostly causatives like *have/let*, perception verbs) are relatively stable across the Germanic languages. By contrast, ECM_2 exhibits considerably more cross-linguistic variation, both in the range of matrix predicates that license the construction and the status of the embedded verb. Across different verb classes—including *say/claim*, *believe*, *consider*, and *expect*—English and Icelandic impose the fewest restrictions, whereas Dutch and German do not permit ECM constructions with any of these predicates. This contrast is illustrated with *say* in Icelandic in (56a) and *expect* in Dutch in (56b). A second status (i.e. a 'to'-marked infinitive) is only possible in Swedish/Norwegian, albeit with restrictions, e.g. non-eventive interpretations (Lødrup 2008).

(56) Icelandic, Dutch

a. Jónas sagði Garp hafa farið í bíó
Jonas said Garpur.ACC have gove to cinema
"Jonas said that Garpur has gone to the cinema." (Wurmbrand & Christopoulos 2020: 392)

b. * dat ik Peter de deur verwachtte te sluiten
that I Peter the door expected to close
"that I expected Peter to lock the door" (Broekhuis 1992: 211)

Other raising predicates involve adjectives or nouns with nonfinite complements, see (57) (Abeillé 2024: 526–527). In a HPSG setting, also different kinds of copula and predicative constructions (resultatives, depictives) have been analyzed as raising configurations (see Müller 2002: Ch. 2).

(57) a. It is *likely* [to rain.]

b. It has a *tendency* [to rain] at this time of year

A puzzling construction that bears certain resemblances to raising is the so-called *tough*-movement construction (Hicks 2009). It involves a nonlocal dependency between the matrix subject and the implicit object position of the infinitive. There is a general consensus that these predicates do not theta-mark their subjects, cf. (58a), yet many specifics have not been settled. Since Chomsky (1977), the standard analysis resorts to an empty *wh*-operator (*Op*) that binds the gap in object position (58b).

(58) a. It is tough to please linguists.

b. Linguists$_i$ are tough [$_{CP}$ *Op*$_i$ to please t$_i$]

A hybrid class in terms of their orientation properties are aspectual or 'semi-modal' verbs ('begin', 'stop', etc.), which can appear in both raising and control constructions, see the contrast between (59a) and (59b). In German, the raising variants are regarded to construe coherently even though the *zu*-marked infinitive can appear in extraposed position. As (59c) shows, these verbs also allow for the third construction (see Reis 2007 and Schallert 2013: Sec. 3.1 for further discussion).

(59) a. It started to rain. (raising)

b. The gardener^i began [PRO^i to mow the lawn.] (control)

c. weil ihm anfing/begann der Kopf zu wackeln
since him started/began the head to totter
"since his head started to totter" (Reis & Sternefeld 2004: 502)

4 Non-sentential (coherent) infinitivals

Sec. 3.1 discussed important test criteria that can be used to determine the differences between coherent (non-clausal) and incoherent (clausal) nonfinite constructions. From a descriptive perspective, it is necessary to assume an intermediate area or allow for problematic cases:

- In the Germanic OV-languages and in German in particular, there are a number of verbs that allow for optional coherent constructions, see Reis & Sternefeld (2004: Sec. 2) and Haider (2010: 278–282). A related phenomenon is the third construction, which Wöllstein-Leisten (2001) considers to be an alternative coherence-forming construction, but which in Dutch tradition is regarded as a sentence-level construction (den Besten & Rutten 1989).
- For causative and perception verbs (that also run under the labels ACI- or ECM-verbs9,[13] it has been argued in some cases that sentence-level embeddings also exist, e.g., based on binding data (Grewendorf 1983); for modal verbs, this has also been done based on temporal reference Hinterhölzl (2006: 131). However, the arguments presented there do not stand up to closer scrutiny (see Schallert 2014b: 245–246, 248–249).

Non-clausal nonfinite construction patterns occur primarily in connection with periphrastic tense, mood, and aspect forms, which may be expanded to include modal verbs. Relevant examples from Dutch (OV) and Swedish (VO) are given in (60). Another important group are causative and perception verb constructions, as illustrated in (61). Note that for these constructions in VO, we have to assume at least VP embedding, typically a TP structure.

(60) Dutch (OV), Swedish (VO):

a. dat de hamster de aardbei zou hebben opgegeten
the hamster should the raspberry should have.PTCP up=eaten
"(that) the hamster is supposed to have eaten the raspberry."

b. (att) hamster-n har ätit.SUP jordgubbe-n
(that) hamster-DEF has eaten raspberry-DEF
"(that) the hamster has eaten the strawberry."

(61) a. att vi hör hamster-n rassla
that we hear hamster-DEF rustle
"that we hear the hamster rustling"

b. dat we de hamster op de bank laten rondrennen
that we the hamster on the sofa let run=around
"(that) we let the hamster run around on the sofa."

4.1 Differences between OV and VO in Germanic

Following the framework developed by Bech (1955), complex predicates can be classified on the basis of two properties: (A) the morphological marking of the dependent verb ('status'), and (B) the dominance relations holding among the members of a verbal chain. As discussed in Sec. 2, Bech (1955: 12)

[13] ACI = 'accusativus cum infinitivo', ECM = 'exceptional case marking' (see below).

distinguishes three types of Status: the simple infinitive (first status), the *zu*-infinitive (second status), and the participle (third status). Verbs and certain other predicates that select nonfinite complements can be classified according to the status they require. We also saw that the dependency relations within the predicate complex are local insofar as one element of the verbal chain selects for the morphosyntactic properties of another one ('status government'), yet it remains unaffected when the governing predicate is itself status-governed. By means of status government, it can also be determined in which hierarchical order verbal elements stand. This order can be marked via indices, with the most deeply embedded verb always bearing the highest index, cf. (62).

(62) dass er alles reparieren$_3$ zu können$_2$ behauptet$_1$
that he everything repair to can claims
"that he claims to be able to repair everything"
(Wöllstein-Leisten 2001: 67)

Note that indexing is usually employed only for the verbs in the right periphery (right sentence bracket, in traditional terms), meaning that the finite verb, which is located in the C-domain in V/1 and V/2 configurations, see (63), is disregarded.[14]

(63) a. Er hat alles reparieren$_3$ zu können$_2$ behauptet$_1$
he has all repair to could claimed
"He claimed to be able to repair everything."

b. Hat er alles reparieren$_3$ zu können$_2$ behauptet$_1$?
has he all repair to could claimed
"Has he claimed to be able to repair everything?"

Order variation of verbal elements in the right periphery is per se a specific property of the Germanic OV-languages (Haider 2010: 33). More specifically, there is a direct connection to the directionality of status government in the verbal complex. While in a VO-language like English, the right-branching serialization (64a) is the only option, an OV-language like German shows the mirror-image pattern (64b).[15] A connection between the setup of complex predicates in terms of branching direction and the basic word order of a language was already established by Greenberg (1963: 85), whose universal 16 states: "In language with dominant order SOV, an inflected auxiliary always follows the main verb."

(64) a. that he might$_1$ have$_2$ been$_3$ seen by John (English)

b. dass er von Johann gesehen$_4$ worden$_3$ sein$_2$ mag$_1$ (German)

4.2 Verb clusters in Continental West Germanic

Only OV, but not VO allows for deviations from the left-branching base order, and almost all Germanic OV-languages deviate from the left-branching base order at least in one construction (Wurmbrand 2004,

14 Some important contributions to verb clusters deviate from this convention (e.g. Dubenion-Smith 2010) but this risks comparability of results with other works.

15 The following definitions hold (cf. Haider 2013: 3–4): Right-branching means that the branching node in a directed graph (tree structure) always follows the non-branching node along the branching line. Left-branching refers to the reverse, i.e. the non-branching node always precedes.

Wurmbrand 2017). Take German Standard German, for instance: There are cases of fronting of the finite auxiliary to the top of the predicate complex, in particular with IPP-constructions but also with *werden* in its temporal/modal uses (cf. Schmid 2005: 40):

(65) dass er ihn die Medizin *hat/wird* trinken lassen
the he him the medicine has/will drink let.INF
"that he has/will let him drink the medicine"

Especially larger verb-clusters show an increasing amount of variation, which may in part be due to the substrate of regional varieties. As an example the variants in (66) can be consulted; experiments with speeded acceptability judgements conducted by Bader & Schmid (2009) showed that some of these variants clearly belong to the repertoire of colloquial German.

(66) Ich bin mir nicht sicher, ob der Fehler…
I am REFL not sure, if the mistake
a. *hätte* bemerkt werden können
had.SBJV noticed become can
b. bemerkt *hätte* werden können
c. bemerkt werden *hätte* können
"I'm not sure if the mistake would have been noticed."

Given the considerable variation in verb clusters across Continental West Germanic, I will focus on the most common patterns, primarily drawing on the excellent survey by Wurmbrand (2017), which also incorporates elicitation data.

- In 2-verb clusters, both the serializations 2-1 (left-branching) and 1-2 (right-branching) are attested in configurations featuring auxiliary-participle (AUX-PTCP) and modal-infinitive (MOD-INF) (Wurmbrand 2017: 4622). While Standard Dutch shows construction-specific variation between both patterns, Standard German and (West) Frisian always exhibit 2-1. German dialects also show both patterns, with subtle construction-specific and diatopic differences (see Weiß & Schwalm 2017: 464 and the references quoted there).
- In 3-verb clusters, all mathematically possible serializations (3! = 6) are actually attested, once again with regional and construction-specific effects. Common orders are 1-3-2, 3-1-2 and the fully inverted 1-2-3, while 2-3-1 is less common. Wurmbrand (2017: 4624), who observes the configurations MOD$_{\text{fin}}$-MOD-V, AUX$_{\text{fin}}$-MOD-V, AUX$_{\text{fin}}$-IPP-INF and AUX$_{\text{fin}}$-PTCP-PTCP, claims that 2-1-3 is absent in her data, yet there is some indication that this serialization does exist in the relevant contexts, albeit under specific information-structural contexts (Schmid & Vogel 2004). In other coherent constructions, this pattern is well-attested, e.g. with causatives embedded under modals in East Hessian dialects (Weiß & Schwalm 2017: 466) or with benefactives, perception verbs and phasals in Alemannic (Lötscher 1978; Schallert 2014b: 201). In Standard German, there is variation between predominant 3-2-1 and 1-3-2 in IPP-constructions and with *werden* (see below). In Standard Dutch, the predominant serialization is 1-2-3, while the configuration MOD$_{\text{fin}}$-AUX-PCTP shows variability between 1-2-3, 3-1-2, and 1-3-2 (ct. also Zwart 2011: Ch. 11). (West) Frisian, by contrast, shows predominantly 3-2-1.

A question that is directly connected to different serialization patterns is the so-called ‘cluster property’ of verbal chains and its grammatical causality. With completely left-branching serializations, the Germanic OV-languages allow no intervening material in the predicate complex (see, among others, Haider 2003: 93); similar effects are also reported for non-Indo-European languages with this base order, for example Japanese or Korean (Sells 1990). Haider takes the following contrasts between English and German as evidence for this systematic difference: Only in the former language (with a right-branching predicate complex) can adverbs be inserted at all verbal positions (67a), whereas for German (with a left-branching predicate complex), this is strictly ungrammatical – indicated by the asterisks at the relevant positions (67b).

(67) a. The new theory certainly may possibly have indeed been badly formulated (English) (Quirk et al. 1985: 495, § 8.20)
b. dass die Theorie wohl tatsächlich schlecht formuliert (*) worden (*) sein (*) mag (German)

In OV-languages, this compactness restriction can only be lifted for construction-specific deviations from the left-branching base order, in particular in partially right-branching configurations. Such structures are called *verb projection raising* (VPR), which has become to be a quasi-descriptive term (see also below). Typical interveners, with language/dialect-specific distribution, can be ordered in an implicational scale like the following (Wurmbrand 2017: 4685): separable particles > ‘low’ (i.e. process-oriented) adverbials, idioms, and bare NPs > indefinite objects and PPs > definite objects. In Standard German, one possible scenario is auxiliary fronting to the top of the predicate complex in IPP-constructions or with *werden* ‘become’. Further evidence for this connection between branching direction and compactness comes from Hungarian, a non-Indo-European language that also shows clustering effects similar to those in German and Dutch (Szendrői & Tóth 2004). Since many of the German dialects show much more verb order variation in the right periphery than the standard language, interactions between branching direction and compactness can be assessed in more detail. Here again, the basic contrasts hold: Left-branching serializations are always compact while their right-branching counterparts are open for nonverbal interveners (cf. the overview by Sapp 2011: 124–128 on several German dialects). Matters are a bit more complicated with so-called mixed serializations, though (see Schallert 2014a: 279–281 for a thorough discussion).

Another problem for this correlation is posed by Dutch (and most of its dialects), since it never allows nonverbal interveners with right-branching serializations, although this is certainly a rather young development.[16] The same applies to Norwegian (VO), which only allows for interveners after the finite verb or directly before the main verb or participle (Nilsen 2003; cf. Haider 2010: 274, Fn. 5). Thus, right-branching is a necessary, yet not a sufficient condition for nonverbal interveners to appear. Note that even in Dutch, evidence for interveners can be found, namely with particle verbs and certain other ‘minimal elements’ (see the synoptic map in Dros-Hendriks 2018: 154, based on SAND-data) and in cases of the so-called ‘third construction’, which occurs with a subset of control verbs and shows clear signs of mono-clausality (‘coherence’ in descriptive terms). Thus, examples like (68) are but an instance of the 2-1-3 pattern that can also be found e.g. with perception verbs in Alemannic (Schallert 2014b: 205–210).

[16] Older stages of Dutch show clear cases of such verb projection raising structures (Hoeksema 1994).

(68) dat Jan zijn huis probeert met winst te verkopen
that Jan his house tries with yield to sell
"that Jan tries to sell his house with yield"
(den Besten & Rutten 1989: 41)

This construction is very common in German (and its dialects) and can also occur with verbs that allow for both raising and control patterns, e.g. *anfangen* 'begin', *drohen* 'threaten', *versprechen* 'promise' (see Wöllstein-Leisten 2001; Reis 2007).

Ever since Evers' (1975) pioneering work, numerous mechanisms have been proposed for dealing with verb order variation and nonverbal interveners in the verbal complex (see Wurmbrand 2017: Sec. 3.2.3 for a thorough overview). Most prominent are the operations *verb raising* (VR) and *verb projection raising* (VPR), which operate over the OV-typical left-branching base order in the verbal complex and lead to adjunction of a verbal head or its projection(s) to a structurally higher position in the verbal complex.[17] As the examples in (69) from Zurich German show, (69a) is the expected base order for an OV-language. In contrast, the serializations (69b) and (69c) are the result of VR and VPR, the latter case leading to nonverbal material ending up in the verbal complex. As noted above, there are relatively big differences between the German (and other Continental West Germanic) dialects as to which of these both operations can apply and/or the relative order of these operations (Wurmbrand 2004; Wurmbrand 2017: Sec. 3.2.2).

(69) Zurich German:

a. das de Hans es huus chaufe wil
that the Hans a house buy wants
"that Hans wants to buy a house"

b. das de Hans es huus e_i wil [chaufe]$_i$

c. das de Hans e_i wil [es huus chaufe]$_i$

4.3 Morphosyntactic alternations

In Continental West Germanic varieties, the most common morphosyntactic effect is the so-called substitute infinitive (infinitivus pro participio) (Schmid 2005; Schallert 2014b, and others). This effect is illustrated with the German and Dutch examples in (70)–(71): In complex perfects, the infinitive occurs instead of the participle, cf. (70); simple perfects, by contrast, always exhibit the expected morphological marking (71).

(70) a. Clara hat das Buch nicht lesen *wollen* | * *gewollt*
Clara has the book not read want.INF wanted.PTCP

b. Clara heeft het boek niet *willen* | *gewild* lezen
Clara has the book bot want.INF | wanted.PTCP read
"Clara didn't want to read the book."

(71) a. Clara hat das nicht *gewollt* | * *wollen*
Clara has that not wanted.PTCP | want.INF

[17] For the sake of illustration, I simplify matters somewhat. Particularly for the analysis of VPR other mechanisms beside adjunction have been proposed, e.g. reanalysis and inversion by Haegeman & van Riemsdijk (1986).

b. Lea heeft dat niet *gewild* | (* *willen*)
Clara has that not wanted.PTCP | want.INF
"Clara didn't want that"

A necessary precondition for IPP is the availability of *ge*-prefixed participles (Schallert 2014a: 259–260), which can also be linked to the OV/VO typology: As Vikner (2001: 77) notes, none of the Germanic VO-languages forms participles this way. However, this is not a sufficient condition since there are OV languages like Frisian or certain Bavarian dialects which do not seem to show IPP (Dammel & Schallert 2021: 214–215; see also Wurmbrand 2017: 4621, 4707).

Within the Continental West Germanic dialect continuum, there is a high amount of variation regarding the IPP effect, the relevant factors being (A) the licit predicate classes and (B) the serialization patterns of the complex verb forms involved. In Standard German, the core classes are causative *lassen* 'let', modals (including *brauchen*[18]), and perception verbs, in dialects and regional varieties a few additional cases are reported, e.g. with *anfangen* 'to start' or *sich trauen* 'dare (REFL)' (Schallert 2014a: 252–253); in Standard Dutch, there is a much bigger nummer of IPP-verbs (Ponten 1973, for instance, lists 33; cf. also the list given in Zwart 2011: 365). While German (including its dialects) shows a clear tendency for the substitute infinitive to occur with verbs that govern the simple infinitive (1st status), this pattern does not hold for Dutch. According to Ponten (1973: 76), roughly two thirds of IPP verbs in Dutch select the 2nd status, including, among others, *beginnen* 'begin', *trachten* 'try', and *gelieven* 'prefer'.[19] It has been repeatedly stated in the literature that IPP is connected to specific reorderings in the verbal complex. As the discussion in Wurmbrand (2017: Sec. 2.2, in particular p. 4618) shows, this correlation does not hold if the wider empirical situation in Continental West Germanic is considered. In particular, IPP is also found in strictly left-branching serializations, which correspond to the base order in OV languages.

A related phenomenon that occurs in the same syntactic configuration is the PPI-effect ('participium pro infinitivo'), mainly known from Continental North Germanic languages like Swedish or Norwegian, but also Frisian (Wiklund 2001; Barbiers et al. 2008a: Map 38a, Barbiers et al. 2008b: 40). In this case, the most deeply embedded, lexical verb bears the participial morphology, see (72). Occasionally, parasitic participial forms like these can also be found in historical/dialectal varieties of German, often in combination with the substitutive infinitive on the superordinate verb (see Jäger 2018: 317–319 and Schallert 2014b: 192–193 for further discussion). While IPP is a speciality of many (Continental) West Germanic OV-languages, PPI occurs both in Germanic VO- (Scandinavian) and OV-languages (Frisian).

(72) Jeg hadde villet lest boka (Norwegian varieties)
I had wanted read.PTCP book-DEF
"I would have wanted to read the book" (Wiklund 2001: 201)

Apart from IPP and PPI, Continental West Germanic shows different other morphological effects in complex perfect constructions, e.g. specialized participles or morphological merger between infinitive and participle (see Höhle 2006; Jäger 2018: 313–319 and Dammel & Schallert 2021: 216–217).

18 From the 19th century onwards, also *brauchen* 'need' is attested with this effect (Lenz 1996; Maché 2019: 195), which is indicative of its morphosyntactic convergence towards the modal verb class.

19 In Standard German, a few show variation between the *zu*-infinitive and the simple infinitive, e.g. *helfen* 'help', *lernen* 'learn' or *brauchen* 'need' (thus patterning with other modals, which typically select for a simple infinitive).

5 Sentential (incoherent) nonfinite constructions

In all Germanic languages, we also find NFCs that have clausal status. There are different empirical diagnostics for clausehood, which have already been addressed in Sec. 3.1, e.g. independent temporal reference, binding of anaphors/reciprocals or extraposition. A morphosyntactic criterion par excellence is the presence of specialized particles in the C-domain. German, for instance, shows *um* in adverbial infinitival clauses (73a), while its Dutch cognate *om* can also appear in complement clauses (73b). In High German dialects like Bavarian and Alemannic, *zum* can be found in both types of clauses (Schallert 2013; Weiß 1998: Ch. 7; Vergeiner & Elspaß 2025).

(73) German, Dutch:

a. Klaus hat sich beeilt, *um* den Zug zu erwischen.
Klaus has REFL hurried, in.order the train to catch
"Klaus hurried in order to catch the train."

b. Klaus probeerde *om* te roken.
Klaus tried COMP to smoke
"Klaus tried to smoke."

English stands out with its prepositional complementizer *for*, which is able to license the external argument of the infinitival clause, cf. (74a). Other examples are less clear cut: The fact that the infinitival marker *att/å* can appear before sentential negation in Swedish or Norwegian (Platzack 1986; Thráinsson 1998: 355–357) may be taken as a diagnostic but this would also mean that in English examples like (74b), *to* would be a complementizer.

(74) English:

a. I want for my hamster to be happy.

b. I told you to not touch it!

Other infinitival constructions, mainly ECM/raising or *tough*-movement constructions have traditionally been analyzed as involving a TP layer, corresponding to their propositional content, yet being 'defective' in terms of their temporal reference potential.

5.1 Syntactic functions

Let us now examine the formal and functional spectrum covered by incoherent infinitivals. They occur in all major syntactic functions, including NP-modifiers ('attributes'). Typical examples from members of both the Germanic OV (German) and VO branches (Norwegian) are given in (75) and (76).

(75) German:

a. Hamster zu beobachten gefällt ihr.
Hamsters to observe pleases her
"It pleases her to observe hamsters."

b. Sie versucht, den Hamster zu beobachten.
she tries the hamster to observe
"She tries to observe the hamster"

c. Sie hat Erdbeeren gekauft, um dem Hamster eine Freude zu machen.
she has raspberries bought COMP the.DAT hamster.DAT a pleasure to make
"She bought some strawberries to make the hamster happy."

(76) Norwegian [Bokmål]):

a. Å se på hamstere gleder henne.

b. Hun prøver å observere hamsteren.

c. Hun kjøpte jordbær for å gjøre hamsteren glad.

We also find oblique case relations such as (77), albeit with noteworthy selectional differences. While German and Dutch require a correlative pronominal element in the matrix clause, this option is not available in the Scandinavian languages.

(77) German, Dutch:

a. Er hat daran gedacht, den Hamster zu füttern
He has CORR thought.PTCP the hamster to feed
"He remembered to feed the hamster"

b. Hij heeft eraan gedacht om de hamster te voeren.
He has CORR thought.textscptcp COMP the hamster to feed

In most cases, thematic selection coincides with directly governed syntactic functions, meaning that complement clauses are linked to subjects/objects, adjunct clauses to adverbials, yet we also find selected adverbials in certain configurations, most notably with (caused) motion verbs (78a) or with degree adjectives (78b):

(78) German:

a. Wir haben ihn nach Bamberg geschickt, um das alte Buch zu kaufen.
we have him.ACC to Bamberg sent COMP the old book to buy
"We sent him to Bamberg in order to buy the old book."

b. Du bist noch nicht groß genug, um eine Flasche Whisky zu kaufen.
you are still not big enough COMP a bottle whisky to buy
"You are not big enough to buy a bottle of whisky."

As already mentioned in Sec. 3.1, German is unique among its relatives in that it allows incoherent infinitive clauses without extraposition, an option that is systematically excluded in Dutch (Haider 2010: 283). However, such cases are restricted to written or more formal registers and seem to be missing in dialects and spoken modalities (Schallert 2013: 133–134):

(79) Hoffentlich hat sie [ihn zu informieren] nicht vergessen
hopefully has she [him to inform] not forgotten
"Hopefully, she has not forgotten to inform him."

5.2 Clause typing

With respect to clause types, English and Afrikaans are notable in that they also allow infinitival embedded *wh*-questions, evidenced by (80) (Harbert 2007: 473; Donaldson 1993: 328). In German and Dutch, this option is highly restricted (Reis 2003: 173–175; Broekhuis & Corver 2015: 604–605), while it is entirely absent in Scandinavian languages. Note that residual examples in German like (81a) always involve a bare infinitive, whereas in Dutch, it is *te*-marked (81b).

(80) Afrikaans, English:

a. Gedurende die vakansie weet min moeders wat om met hul kinders te doen.
during the holidays know few mothers what COMP with their children to do
"During the holidays few mothers know what to do with their children"

b. I just don't know what to do with myself (Dusty Springfield)

(81) German, Dutch:

a. Sie weiß nicht so recht, was tun damit.
she knows no so ADV what do with.it
"She's not quite sure what to do with it."

b. Jan vroeg me hoe die auto te repareren.
Jan asked me how that car to repair
"Jan asked me how to repair that car." (Broekhuis & Corver 2015: 605)

Concerning relatives, English once again stands out in that it allows this option in different contexts, e.g. with object relative clauses (82a), yet an overt relative pronoun is only licit with prepositional objects and pied-piping (82b) (see Harbert 2007: 471).

(82) a. I found a man (*who) to hire.

b. I found a topic about which to write

What is somewhat unclear is whether there are parallels in the other Germanic languages. Thráinsson (2007: 412, 430) points to examples like (83) where a modification relation seems to exist between the *að*-infinitive and the NPs 'wax' while the preposition *með* 'with' is stranded; furthermore, a relative pronoun is always missing. Similar cases are also reported from Norwegian (K. K. Christensen 1984), yet it is more plausible to analyze these configurations as a subtype of purpose clauses (see Harbert 2007: 472–473 for discussion). Note, however, that the preposition *til* in these examples acts more like a complementizer and not like a lexical preposition in the sense of 'for' or 'to'.

(83) Þetta er bón [til að bóna bíla með]
this is wax for to polish cars with

For German, apparent examples for relative clauses involve *zu*-marked infinitives and are only licit with (optionally) incoherent verbs like *sich trauen* 'dare (REFL)' (84a) but not with coherently constructing modal verbs like *wollen* 'want' (84b). There are different analyses for this pattern but there seems to be a consensus that we are dealing with non-canonical relative clauses (see Müller 1999: Ch. 10.7 and Haider 2010: 296–297 for further discussion).

(84) German:

a. die Ratten, [die zu fangen] sich niemand traute
the rats which to catch REFL nobody dared
"the rats, which nobody dared to catch"

b. * die Ratten, [die fangen] niemand wollte
the rats which catch nobody wanted
"the rats, which no one wanted to catch"

A peculiarity of German are root infinitivals like (85), which come in different illocutionary flavors (see Fries 1983 and Gärtner 2013 for an overview), mostly non-canonical imperatives/adhortatives, and questions. With the exception of certain expressives (86), they only feature infinitives (1st status) or participles (3rd status). In the other Germanic languages, this option seems to be much more restricted (but see Platzack 2017 on Swedish).

(85) German (Fries 1983: 1):

a. Bitte nicht auf den Boden spucken!
please not on the floor spit
"Please do not spit on the floor"

b. Wohin sich noch wenden?
where REFL yet turn
"Where else to turn?"

c. Aufgestanden!
up=stood.PTCP
"Up you get!"

(86) (Ein Wunderkind!) So schön Geige zu spielen!
(A child prodigy!) How wonderful to play the violin like that! (Gärtner 2013: 205)

For root-*wh* questions, Reis (2003: 189–191) points to the fact that their illocutionary potential is defined by their function as self-directed, deliberative 'uncertainty' questions that express doubt about the existence of a clear answer, as evidenced by the contrast between a normal, information-seeking *wh*-question (87a) and its infinitival counterpart in (87b); in English, such structures are generally highly idiomatized and mostly appear with the wh-item *why* (cf. Reis 2003: 157).

(87) a. Wen soll man Ihrer Meinung nach fragen?
"Who should one ask in your opinion?"

b. ?* Wen Ihrer Meinung nach fragen?
(Reis 2003: 190)

The standard approach for explaining the diachronic genesis of patterns like these is *insubordination* (Evans 2007), i.e. independent use of (formally) dependent structures with conventionalized ellipsis of the main clause, yet in contrast to respective finite examples like (88), root infinitivals seem to involve additional elliptical steps (see Evans 2007: 384 on reduced insubordinated clauses in Japanese). A diachronic survey on these structures is still missing.

(88) Dass der seinem Hund ein BUCH vorliest!
that DEM his.DAT dog.DAT a book to.reads
"Can you believe he's reading a BOOK to his dog!" (D'Avis 2013: 178)

6 Summary and Outlook

This chapter has provided a descriptive overview of nonfinite constructions (NFCs) across the Germanic language family, highlighting both their underlying structural unity and their significant empirical diversity. As demonstrated, the behavior of NFCs is closely tied to broader typological features, most notably the basic word order within the VP, which creates a primary divide between the OV type (German, Dutch, etc.) and the VO type (English, Scandinavian). Several key insights have emerged from this survey:

- While all Germanic languages employ nonfinite forms—categorized here using Bech's (1955) status theory—they show varying degrees of morphological reduction. Conservative systems like German and Icelandic maintain robust person/number distinctions in finite forms that contrast sharply with their nonfinite counterparts, whereas languages like English and Afrikaans have seen significant syncretism or loss of these markers.
- The distinction between coherence/incoherence on the one hand and raising/control on the other hand (*orientation* in Bech's 1955 terminology) remains a fundamental tool for understanding structural differences between different types of nonfinite embeddings and thematic and anaphoric dependencies between matrix and embedded predicates.
- A defining feature of Continental West Germanic (OV) is the ability to form verb clusters where in certain configurations, strict compactness holds while in others, nonverbal interveners are licit—nonetheless, all relevant configurations show no signs of clausal behavior. While in VO, verbal complexes usually do not show compactness, the hallmark criterion is that only OV shows verb order variation in these configurations, while in VO, the order ist strictly ascending (or right-branching). Across Germanic, different morphological effects can be observed in coherent constructions, among them the 'substitute infinitive' (infinitivus pro participio = IPP) in Continental West-Germanic or its converse, 'substitute participles' (participium pro infinitivo = PPI), which also occurs in (Continental) North Germanic.
- There is robust evidence for sentential (incoherent) NFCs across Germanic. They appear in the core syntactic functions and can also display specific complementizers (e.g. *um/om* in German and Dutch). Other types like infinitival *wh*-questions or root infinitivals seem to be more language-specific.

Despite the depth of existing research, several open questions remain. The diachronic emergence of specialized forms, such as progressives with *am/aan het* or the English gerund, raise questions about the categorial status of the respective forms. The in principle clear division between ± coherent structures raises many questions as to which predicate classes allow for which type. The same applies to the differences between raising and control predicates and their semantic basis. Additionally, the high degree of dialectal variation of serialization patterns and ensuing morphological effects like IPP raise important questions about the interaction between morphology, syntax, and processing in the Germanic verbal complex. Let me close with a nice quote by H. P. Lovecraft (according to ChatGPT):

> In the abyssal depths of the Germanic tongues, the nonfinite forms endure like forgotten runes: the infinitive, the participle, and the verb-shadows unbound by time and mortal will. They are the echoes of ancient speech, lingering between being and becoming, where the dead voices of forgotten ages still whisper.

Morphosyntactic glosses 1, 2, 3 = person features; ACC = accusative; ADV = adverb; AUX = auxiliary; COMP = complementizer; CORR = correlative pronoun; DAT = dative; DEF = definiteness; INF = infinitive; IM = infinitive marker; IPP = infinitivus pro participio; MOD = modal verb; NOM = nominative; PASS = passive; PL = plural; PPI = participium pro infinitivo; PRS = present; PST = preterite; *pro* = null pronoun; PRO = (silent) infinitival subject; PTCP = past participle; REFL = reflexive; SG = singular; SUP = supine; VC = verbal complex. The glosses follow the *Leipzig Glossing Rules:* https://www.eva.mpg.de/lingua/pdf/Glossing-Rules.pdf (accessed on 31 July 2026).

Use of generative AI I used OpenAI's ChatGPT and Google's Gemini Notebook for stylistic revisions and to generate short summaries. I manually corrected or revised all of the resulting output. Various language examples were generated using ChatGPT. I consulted native speakers for checking the accuracy of these examples. For all remaining shortcomings, I take the sole responsibility.

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